

Engineering

Writing-up your final project

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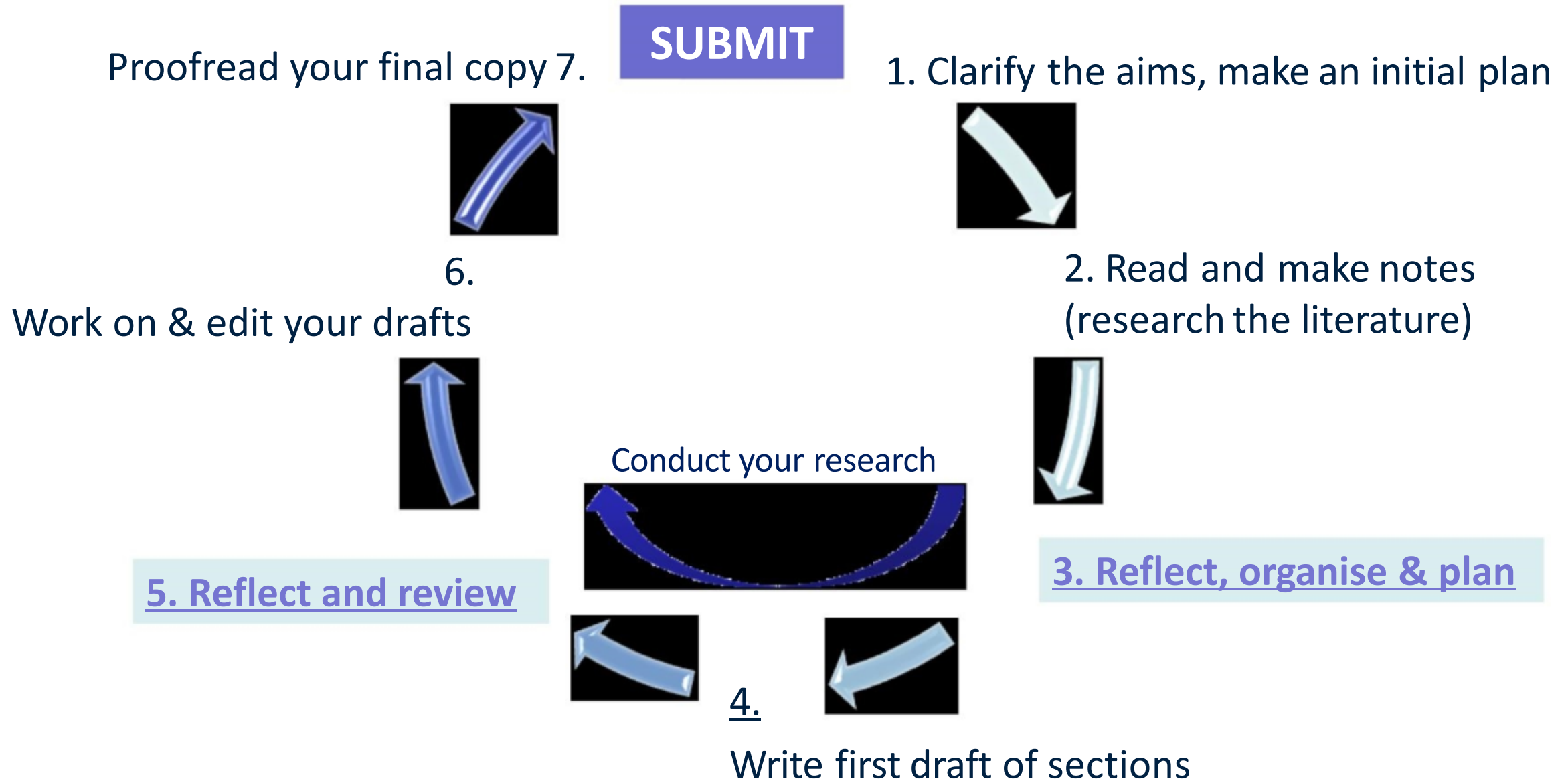
Session overview

- ❑ Further study skills advice and resources
- ❑ Planning and organising your report writing
- ❑ Analysing extracts from example projects
- ❑ Writing Skills:

Clear communication, synthesis of the literature, critical analysis



Writing-up your project: Planning and organising



Writing-up your project: structure

Title page
Table of contents
1. Abstract
2. Introduction and background
3. Literature review
4. Methodology and theory
5. Experimental work / analytical investigation / design
6. Experimental or analytical results / description of final constructed product
7. Discussion and conclusions
8. Project Management, consideration of relevant sustainability and health and safety issues where appropriate
References

Writing skills

What do we mean by 'critical appraisal'?

- ☐ Taking an open, questioning and objective view when considering evidence.
- ☐ Viewing a subject from more than one perspective, i.e. considering evidence which may be contradictory or which does not necessarily support your line of reasoning.
- ☐ Evaluating strengths and weaknesses of research studies, how the issues impact on the findings and perhaps making suggestions for changes or improvements.

Writing skills

What do we mean by 'critical appraisal'?...continued

- ☐ Addressing the 'how' and 'why' as well as the 'what', i.e. not just describing.
- ☐ Discussing the potential limitations of your study / results and making suggestions for changes / improvements and further research.
- ☐ Comparing your results to previous research / studies and drawing some conclusions about your aims / research question(s).
- ☐ Considering the implications of what you have written. What impact does the evidence and your conclusions have on the research field or on other contexts.

Analysing example projects

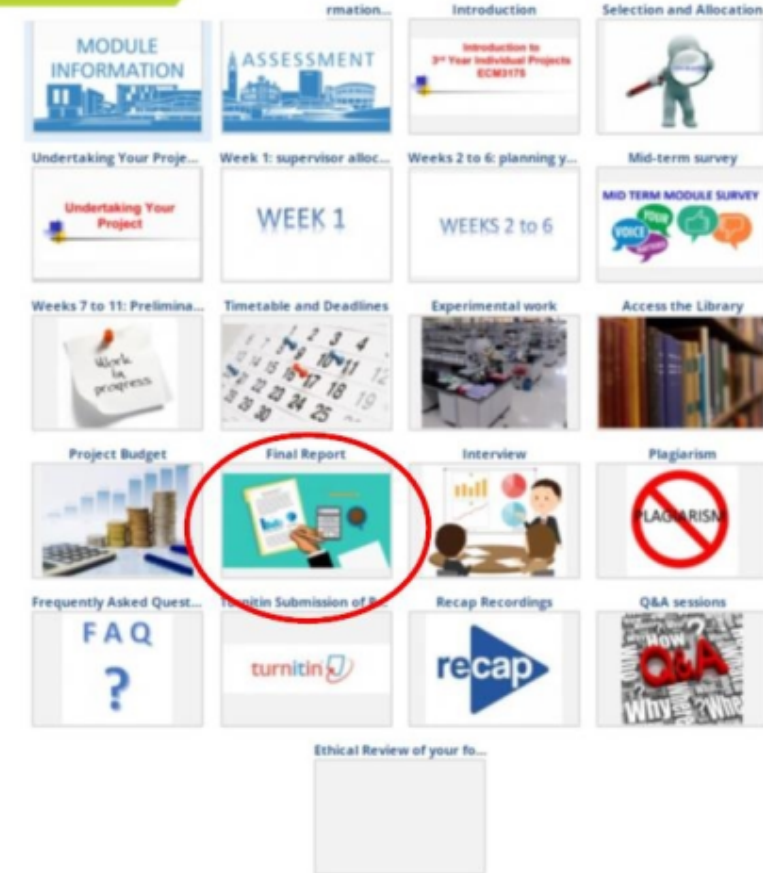
Independent Task 1:

- Go on ELE on the “Final Report” folder.
- At the bottom of the page you will find 2 links to examples of final reports from previous years.
- Select 1 of the report examples.
- Look through the following sections from the example final report that you selected.

- ☐ Abstract
- ☐ Introduction

Final Reports from past years

 Final reports examples



Use the task sheet provided on the next slide to help you identify how well the report meets the marking criteria for each section.

Use this task sheet to help you identify how well the report meets the marking criteria for each section

Section		NOTES / COMMENTS / QUESTIONS
Abstract	• Background to project • Project aim • Methodologies used • Main outcomes • Conclusions	
Introduction and background	• Background to project • What • Why • How • Aims • Objectives	

Analysing example projects

Independent Task 2:

Now look through the following sections from the example final report that you selected

- ☐ Literature review
- ☐ Discussion / Conclusion

Final Reports from past years

 Final reports examples



Use the task sheet provided on the next slide to help you identify how well this report meets the marking criteria for each section.

Use this task sheet to help you identify how well each piece of work meets the project criteria for each section

Literature review	• What has been done before • How it shapes your project • Good referencing	
Discussion and conclusions	• Description of what has been learnt from experimental/analytical work or design • Distillation of important findings and their significance • Specific observations to inform future research	

Writing skills

Use appropriate sources!

Select appropriate and relevant research literature

Avoid overuse of web articles / non peer-reviewed sources e.g.

References (from one of the Examples)

[1] Thermopedia. (2011) *Shell and Tube Heat Exchangers*. [Online] Available from: <http://www.thermopedia.com/content/1121/>

[4] Slideshare. (2014) *Types of Heat Exchangers*. [Online] Available from: <http://www.slideshare.net/AjaykumarAsodariya/heat-exchanger-41154656>

[13] Engineering Toolbox. (2015) *Heat Transfer Fluids*. [Online] Available from: http://www.engineeringtoolbox.com/overall-heat-transfer-coefficient-d_434.html

[15] Wikipedia. (2016) *Porosity*. [Online] Available from: <https://en.wikipedia.org/wiki/Porosity>

Writing skills

(Lit Review Example)

Paragraph structure:

Opening topic sentence

State the point you want to make

Justification & Support

Unpack the topic sentence - explain and analyse the issue

Synthesise evidence and examples from the literature / research studies – compare and critically appraise the findings

Implications

So what does this mean? Conclude the paragraph.

Writing skills

(Lit Review Example)

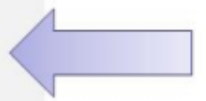
Integrating the literature into your writing

Cost-benefit case studies comparing AM processes to conventional methods, in the form of breakeven analysis's; have verified that for low production volumes AM can be economically feasible [15-19, 26]. Initially, Hopkinson and Dickens found AM to be more suitable for low production volumes, assuming the cost per AM part to be independent of the number of parts [15, 16]. Ruffo et al. developed upon this; showing that the cost per AM part decreases with an increasing number of parts, based upon the LS process [17]. Step changes in cost per part were observed from their model, when increasing the number of parts required the use of another layer or machine. Atzeni and Salmi's group have also undertaken breakeven analysis case studies, but re-designed the parts to utilise AM's advantages [18, 19]. This showed that AM can be economically feasible for higher production volumes, using polymers [18] or metals [19]...

**Good
opening
sentence**



**Synthesis of
research
findings into a
new
interpretation**



Writing skills

(Lit Review Example)

Integrating the literature & evidence of critical appraisal

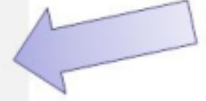
...However, these studies were based upon the Italian economy; so may not be entirely applicable to CCM. The most recent breakeven analysis by Lindemann et al., considered lifecycle costs; showing the step changes from Ruffo et al.'s model to be negligible [26].

Due to the rapid development of AM in recent years [27], these processes may have become more economically feasible since these cost-benefit studies, dated from 2001-2012, were undertaken.

**Critical appraisal:
pointing out strengths
and weakness of studies**



**Good concluding
sentence**



Discussion - evidence of critical appraisal

Elimination of Retaining Wire

...

Nonetheless, the method used to estimate the UTS of the retaining wire, at the strain rate at launch, provided only a rather crude approximation. As the strain rate of the retaining wire at launch, was itself estimated using equation 3.14, to be orders of magnitude greater than the capabilities of the steady tensile testing. Hence, the results were extrapolated over a large range, and it is unlikely that the same relationship between UTS and strain rate will hold over this range of values. Hence, any design emerging from this research, must be exhaustively tested to ensure functional consistency of fin deployment.

Evidence of critical appraisal:

discussing the limitations of the study / results

and making suggestions about what would need to be done

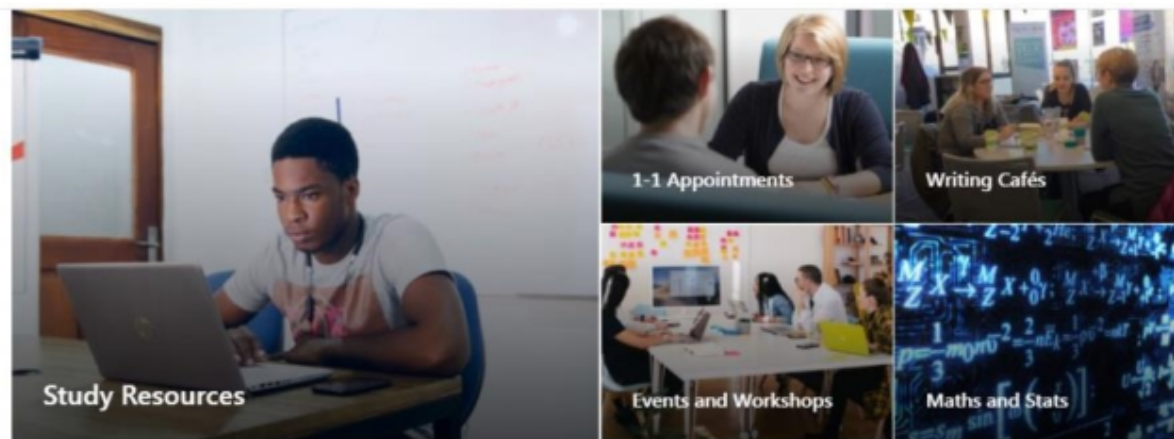


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- access online and on-campus individual appointments
- a drop-in writing café
- study skills resources

Thank you!